

Vijay Karthick R

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Biomedical Engineer | Medical Device & Instrumentation | Signal Processing

Professional Summary

Biomedical Engineering graduate with hands-on experience in medical device design, PCB development, and biosignal-based systems. Hospital-trained in biomedical equipment operation and maintenance, with working knowledge of MATLAB for signal processing and analysis. Combines hardware design (Arduino, KiCad) with front-end visualization (React) to build complete patient-monitoring solutions. Strong interest in medical device engineering, biomedical instrumentation, and healthcare technology.

Technical Skills

Biomedical:	Biomedical Instrumentation, Signal Processing, Medical Equipment Operation & Maintenance, PCB Design
Software & Languages:	MATLAB, Java, C, HTML, CSS, React.js, Node.js, Express.js, MongoDB, SQL
Tools & Platforms:	Arduino IDE, KiCad, Visual Studio, Postman
Concepts:	Data Structures, Algorithms, Object-Oriented Programming (OOP)
Soft Skills:	Leadership, Time Management, Problem Solving

Projects

AI-Based Spirometer with Website Integration	Jan 2024 – Mar 2024
<i>PCB Designer, Front-End Developer</i>	<i>KiCad, Arduino IDE, React.js, MongoDB, Postman</i>
- Built a smart spirometer integrated with a React-based web dashboard to monitor patient lung health - Designed PCB and enabled real-time report generation for doctors using connected sensors and front-end visualization	
Smart Parkinson's Spoon	Apr 2023 – May 2024
<i>PCB Designer</i>	<i>KiCad, Arduino IDE</i>
- Created a gimbal-based spoon using servo motors to stabilize tremors for Parkinson's patients - Designed and fabricated PCB for the project, improving daily usability for patients with motor disorders	

Experience

Project Intern – MERN Powered Hospital Digital Twin Dashboard	Dec 2025 – Mar 2026
<i>FortuMars AI Technologies and Business Solutions Pvt. Ltd.</i>	<i>Coimbatore, Tamil Nadu</i>
- Designed and developed a web-based digital twin dashboard for hospital operations, visualizing patient flow, resource utilization, bed occupancy, and operational metrics - Built the system using the MERN stack under industry guidance, applying modern development tools and methodologies to a healthcare operations use case	
Bio-Medical Equipment Internship Trainee	Aug 14, 2025 – Aug 18, 2025
<i>Aishwaryam Speciality Hospital</i>	<i>Salem, Tamil Nadu</i>
- Completed a 5-day hands-on training program on hospital biomedical equipment operation and maintenance - Gained exposure to real-world medical device workflows, equipment handling protocols, and clinical equipment environments	

Education

Bannari Amman Institute of Technology	Nov 2022 – Apr 2026
B.Tech. Biomedical Engineering, CGPA: 7.37	Tamil Nadu, India
AKV Public School (CBSE)	May 2020 – Apr 2021
12th Grade – 61%	Tamil Nadu, India
AKV Public School (CBSE)	May 2018 – Apr 2019
10th Grade – 73.6%	Tamil Nadu, India

Certifications

- Bio-Medical Equipment Internship Training – Aishwaryam Speciality Hospital (2025)
- Getting Started with Microsoft Excel – Coursera

Areas of Interest

- Biomedical Instrumentation & Signal Processing
- Medical Device Design
- PCB Designing
- Front-End Development